

London Planetree

Platanus x acerifolia

Species background: A hybrid between the American sycamore and the Oriental planetree, first grown in Europe in the 1600s -- possibly an accident of two trees planted near each other in a garden. Bark exfoliates in irregular puzzle-piece patches of cream, olive, and gray, exposing smooth new bark underneath -- this is the single easiest way to identify it at any time of year.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

London Planetree: Identify London planetree by its patchwork, peeling bark and describe why trees shed bark.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons, a real bark sample or photo if available.

Teacher background

London planetree's bark can't stretch as fast as the trunk grows underneath, so it sheds old bark in patches instead -- like outgrowing old shoes. This leaves behind a cream, olive, and gray puzzle-piece pattern.

Procedure

- 1 Engage: Ask -- have you ever outgrown a pair of shoes? What happens to trees that outgrow their bark?
- 2 Explore: Look at real bark (or a photo) and count how many different colors of patches you can find.
- 3 Explain: Explain that the trunk grows faster than the bark can stretch, so it cracks and peels off in patches.
- 4 Art: Do a bark rubbing with paper and the side of a crayon to catch the patchwork colors.
- 5 Evaluate: Student can point out at least two bark colors and explain in their own words why the bark peels.

Discussion questions

What other things in nature shed an old layer to make room for growth?

Assessment

Student names two bark colors and gives a simple, correct reason bark peels.

Extension

Find another tree with peeling bark on your walk and compare its pattern to the planetree's.

London Planetree: Measure a peeling bark patch and explain how bark physiology relates to tree growth.

Standards connection: NGSS 4-LS1-1; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, ruler, crayons.

Teacher background

Bark can't stretch as fast as the trunk grows underneath, so London planetree sheds it in patches instead of expanding smoothly. This species is one of the most-planted street trees on Earth because it tolerates compacted soil, drought, and air pollution better than almost any other large shade tree.

Procedure

- 1 Engage: Ask why a tree planted in a busy city sidewalk might need to be extra tough.
- 2 Explore: Measure the size of one peeling bark patch (real or pictured) with a ruler.
- 3 Explain: Connect bark shedding to trunk growth, and introduce why this toughness makes it a popular city tree.
- 4 Art: Bark-rub at least two patches with crayon and compare the colors and shapes that show up.
- 5 Evaluate: Student records one measurement and writes one sentence connecting bark shedding to trunk growth.

Discussion questions

Why might a city plant one tough, adaptable species on so many blocks?

Assessment

Accurate measurement plus a correct one-sentence explanation of bark shedding.

Extension

Compare planetree bark patches to the bark of a nearby different species.

London Planetree: Explain bark physiology (cambium growth) and connect it to this tree's role in urban pollution capture.

Standards connection: NGSS MS-LS1-1; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, bark-rubbing materials (paper, crayons).

Teacher background

The trunk widens via a growth layer called the cambium; inflexible bark cracks and sheds to keep up -- which also helps shed pollution particles that settle on the surface. This tree tolerates compacted soil, drought, and air pollution better than almost any other large shade tree, which is why it's one of the most-planted street trees on Earth.

Procedure

- 1 Engage: Why might shedding bark help a tree survive in a polluted city, beyond just making room to grow?
- 2 Explore: Research the cambium layer and how trunk diameter growth works.
- 3 Explain: Connect cambium growth, bark shedding, and pollution-particle removal into one explanation.
- 4 Art: Make a bark-rubbing collage from at least 4 patches, then sketch how the pattern might work as camouflage against a brick wall or stormy sky.
- 5 Evaluate: Write a short paragraph explaining the cambium-to-shedding-to-pollution chain in your own words.

Discussion questions

Could a tree with inflexible, non-shedding bark survive as well on a busy city block? Why or why not?

Assessment

Paragraph correctly connects cambium growth, bark shedding, and pollution tolerance.

Extension

Research one other physical adaptation city trees use to survive pollution or compacted soil.

London Planetree: Evaluate this tree's role in particulate interception and produce a technical bark study.

Standards connection: NGSS HS-LS1-3 / HS-ESS3-4; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, measuring tape, graphite pencils.

Teacher background

City trees intercept airborne particulates on leaf and bark surfaces. This hybrid -- a cross between American sycamore and Oriental planetree first grown in Europe in the 1600s -- became one of the most-planted urban trees on Earth partly because of this pollution tolerance.

Procedure

- 1 Engage: How might scientists measure a single tree's contribution to local air quality?
- 2 Explore: Measure this tree's trunk circumference and estimate canopy area using a simple geometric approximation.
- 3 Explain: Research how canopy size correlates with pollution-interception capacity in urban forestry studies.
- 4 Art: Sketch a cross-section of one peeling bark patch like a geologist logging rock strata -- layer by layer, oldest to newest -- and label which layer is actively alive.
- 5 Evaluate: Submit the trunk/canopy measurements, a short research summary, and the labeled bark sketch.

Discussion questions

Is planting one highly pollution-tolerant hybrid species everywhere a good long-term urban forestry strategy? What's the tradeoff?

Assessment

Rubric covering measurement accuracy, research summary, and illustration labeling.

Extension

Compare this species' pollution-tolerance strategy to green ash's vulnerability to emerald ash borer -- what does each case suggest about single-species reliance in cities?

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.